# Adjusting for Survey Non-Response Bias Using Weights in R

John Naughton 11/30/23

Aka “why won’t she text me back?”

# Theory Info

- Example by Vinh Nguyen on October 5th, 2023
- Published in CRAN (Comprehensive R Archive Network) , titled "Example 2: Adjusting for Survey Non- Response Using Weights"

# What is Survey Non-Response Bias?

- Surveys collect and measure valuable information, and response rate is typically very low.
- When only a subset of individuals respond, surveys can suffer from non-response bias because responders may by systematically different than non responders. (for example, responders to an email are probably younger and more tech savvy, telephone responders are more likely to be old and luddites.)
- It's customary to quantify the survey response rate to communicate the possibility of non-response bias.

# How to Minimalize Non-Response Bias

- First Choice: Increase Response Rate (with incentives, change method, etc.)
- Second Choice: Build model for survey response, produce survey weights, re-weight observations. Remember, a weighted analysis is useful ONLY if the analysis has access to variables that relate to decisions around participation.
  - Primary Method is to build a parametric regression model that has transparent relationships between the covariates.
  - Mr. Nguyen suggests an alternative, and R package "twang" that uses gradient boosted trees, a machine learning algorithm, to derive survey weights. This is less useful for drawing inference on survey response (because no covariates given) but better at deriving the non-response weights.

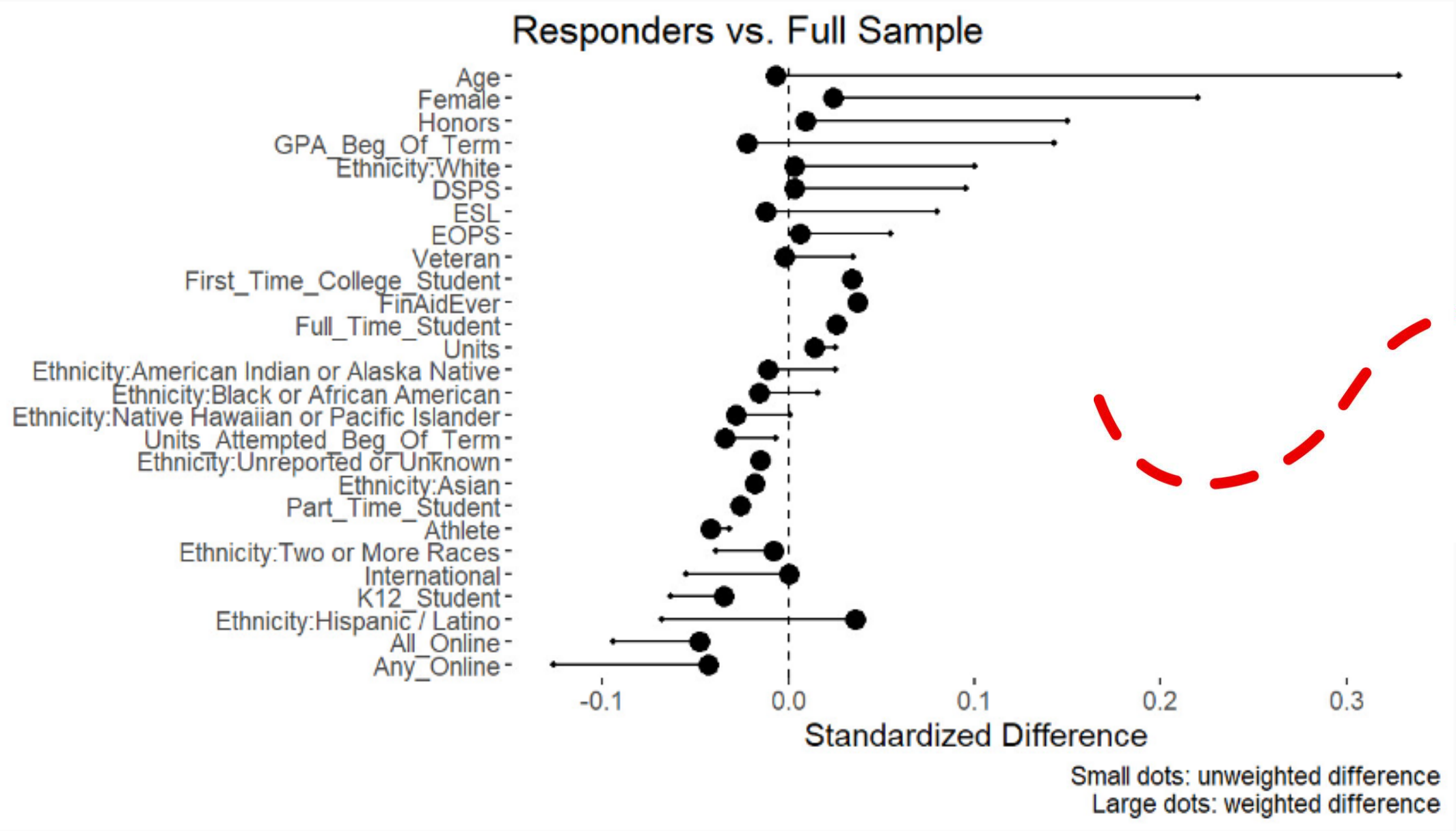
# Methodology and Code

- We did an example based on survey data in R based on college demographics, 28 variables although many were redundant, N=12379 n=1423 response rate= 11.5%
- Remember, the desired results of the weights are that our weighted observations from the sample are similar to the (theoretical) full sample that includes responders and non responders.
- The "twang" process begins by constructing a synthetic data set with weight variables (derived from a simpler parametric model, not too complicated in R) to "trick" the functions in the twang package to compare the full synthetic sample to the actual responder's data. The two data sets are COMBINED, and two new synthetic treatment variables are created. The trt_trick=1 for synthetic data set and trt_trick=0 for the responder data set, and weight_trick=1 for the synthetic data set and weight_trick= the weight we are going to optimize.

# Some sample code

```
d_demo_summ_trick <- bind_rows(
  bal.table(survey_trick_summ)[[1]] %>%
    mutate(Comparison_Type='Unadj.', Variable=rownames(bal.table(ps_fit)$unw))
  , bal.table(survey_trick_summ)[[2]] %>%
    mutate(Comparison_Type='Adj.', Variable=rownames(bal.table(ps_fit)$es.mean.ATE))
  ) %>%
  # filter(!(Variable %in% c('NA'))) %>%
  filter(!is.na(std.eff.sz)) %>%
  select(Comparison_Type, Variable, std.eff.sz)
```

# Resulting "twang" weights

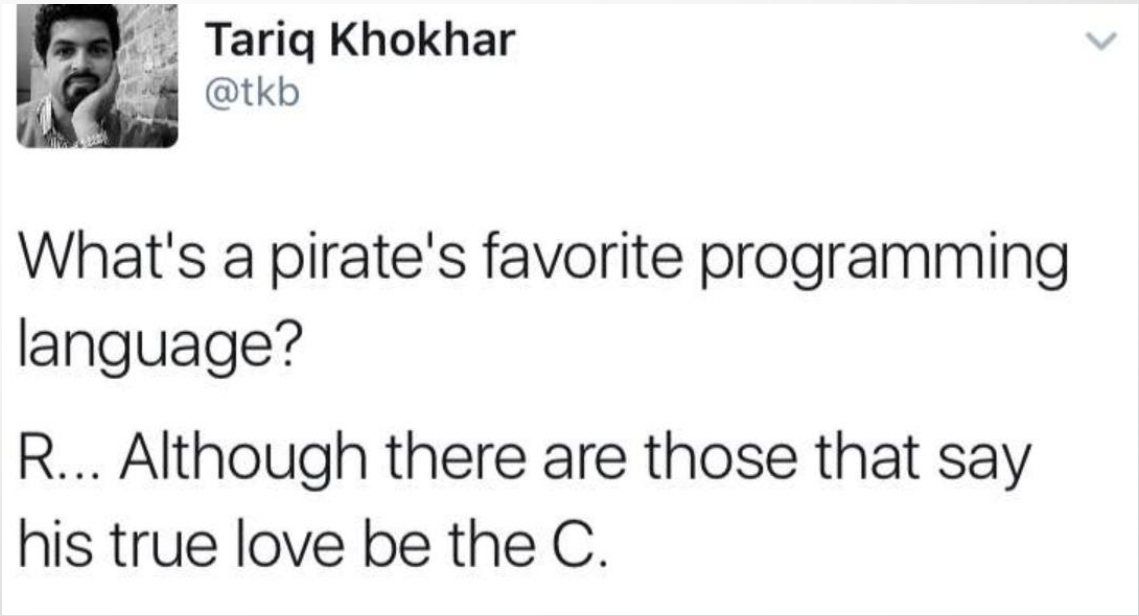


# Interpreting the Results/Recap

Variables which were overrepresented and had to be strongly weighted down: Higher Age, Female Gender, In the Honors Program,  High GPA, White, ESL

- Variables underrepresented and weighted higher: All or Any Online Students, Hispanic/Latino, International Students, K-12 taking college class.
- The use of twang survey weights reduced the imbalance across ALL variables, although some far more than others.
- Assuming the survey response model reasonably captures participants' decision to respond or not, the results can be generalized for the target population.
- We note that the resulting weights don't differ much from the parametric model, but in a more uneven responding scenario the "twang" machine learning approach would pay higher dividends.

# Thank you all!




- "5/5 and 5/5, great presentation" –John Naughton